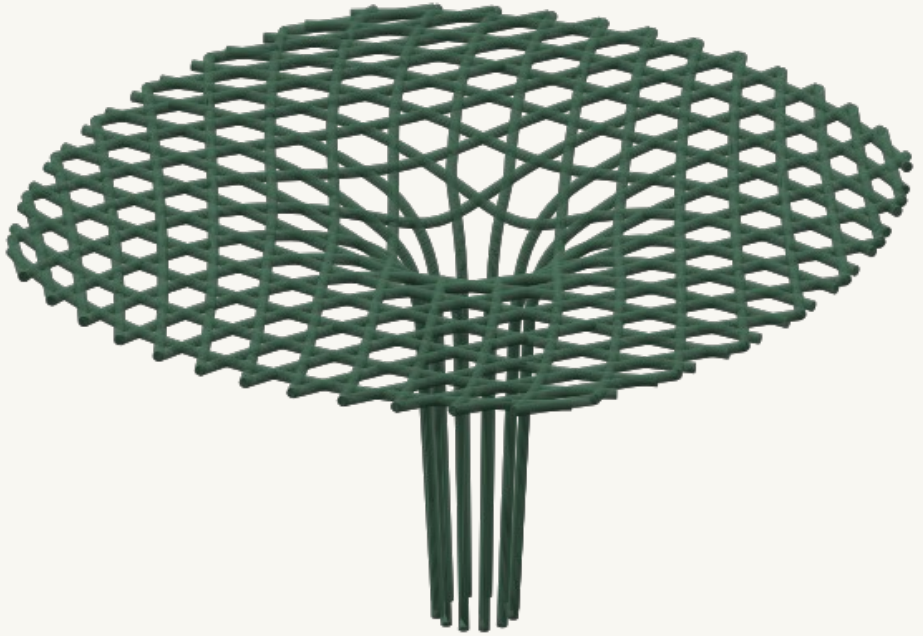
VineWeave Canopy

A tree-like climbing-plant support for pedestrian shade

Parametric design · Planting · Shade performance



TEAM & RESPONSIBILITIES

臧唯羽

Form & parametric modelling

林新淇

Materials & construction

虞寒哲

Shade performance & application

徐世程

PPT production & integration

陈淼

English presentation & Q&A

Translate the brief into five design decisions

A public shade facility must connect environmental intent with a buildable support.

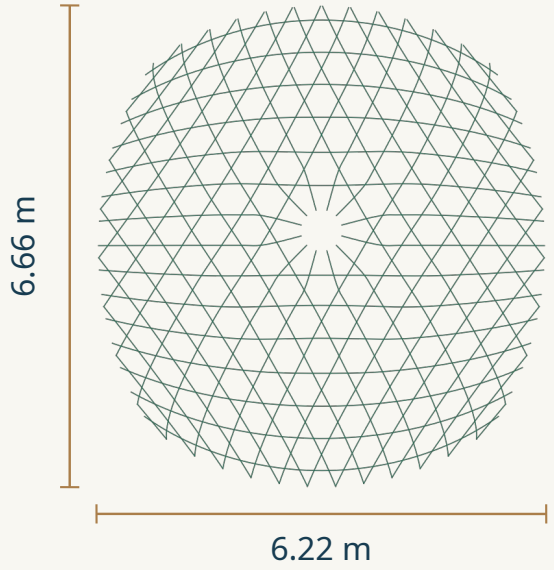
COURSE REQUIREMENT	OUR RESPONSE	WORK PRODUCED
Structure and materials	Steel supports + proposed bamboo crown	Material screen, member schedule and node concept
Geometric parameters	Control crown width, height and corners	Rhino/GH geometry, plans and parameter models
Shade and airflow	Broad crown + planting + open sides	Shadow projection; conceptual airflow analysis
Plant integration	Screen climbers and organise maintenance	Candidate screen, 2+1 planting proposal, cover scenarios
Assembly and use	Prefabricate, assemble and check occupied-hour shade	Build-up sequence, budget and rest-node application

The design is assessed as a system—not as an isolated shape.

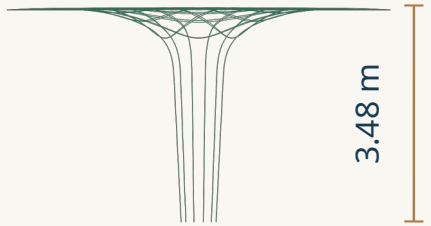
Shade/greening mechanisms: [1,2]. Project requirements and proposed strategy: P4.

Define the final geometry—and how it is measured

Dimensioned plan/elevation views; a final parameter schedule, separate from sensitivity settings.



PLAN



ELEVATION

Curved support inclination is defined by its endpoint chord, not a local tangent.

FINAL PARAMETER	VALUE / DEFINITION
Overall height	≈3.48 m
Plan extents	≈6.22 × 6.66 m
Maximum horizontal span	≈6.91 m, exterior mesh
Support inclination*	45.9–50.7° to horizontal
Member exterior diameter	80 mm
Corner / expansion	0.85 / 1.08

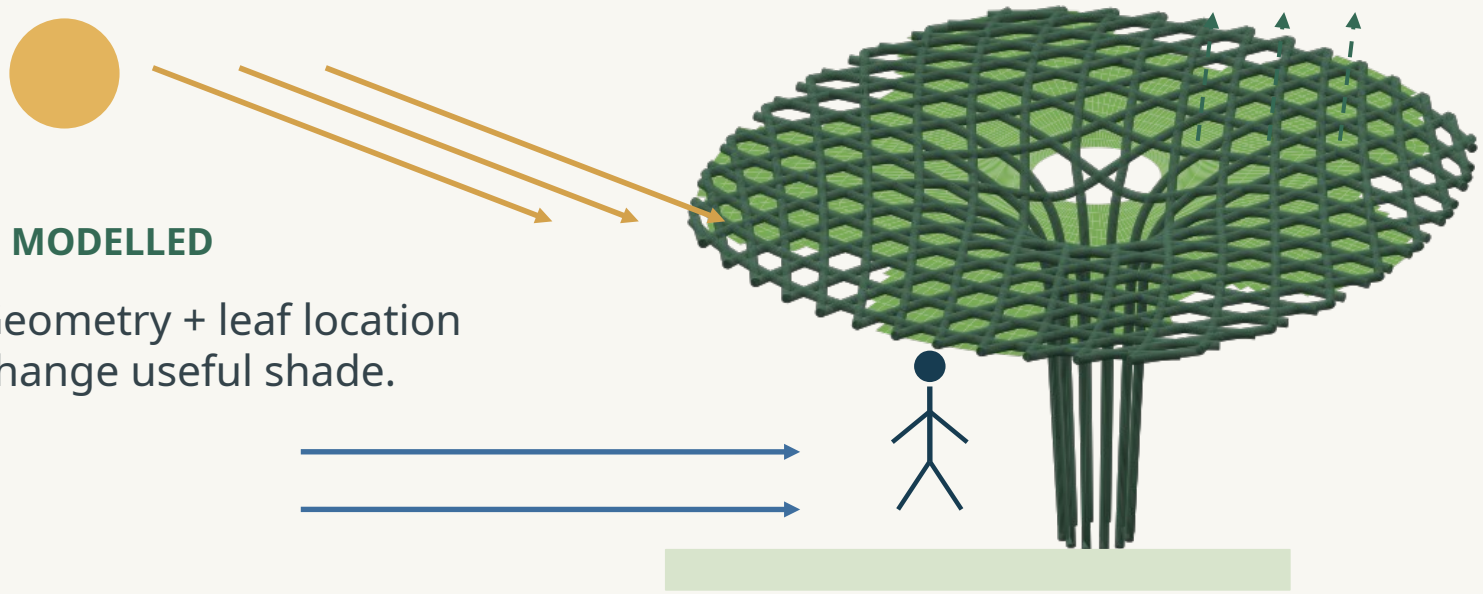
*Angle of each ground-reaching centre-line endpoint chord. Not a uniform member slope.

Model records P1 and supplemental geometric check. Centre-line views; exterior dimensions rounded. Drawing sheets accompany the PPT.

How the canopy can improve pedestrian comfort

Three physical pathways, with the strength of evidence shown next to each.

01 DIRECT SUN



MODELLED

Geometry + leaf location change useful shade.

03 PLANT WATER

Evapotranspiration may exchange latent heat; it needs healthy, watered plants.

02 AIRFLOW

Open sides retain possible air paths.
Dense foliage can also add resistance.

01 SHADE → LOWER EXPOSURE

Direct sun on people and ground

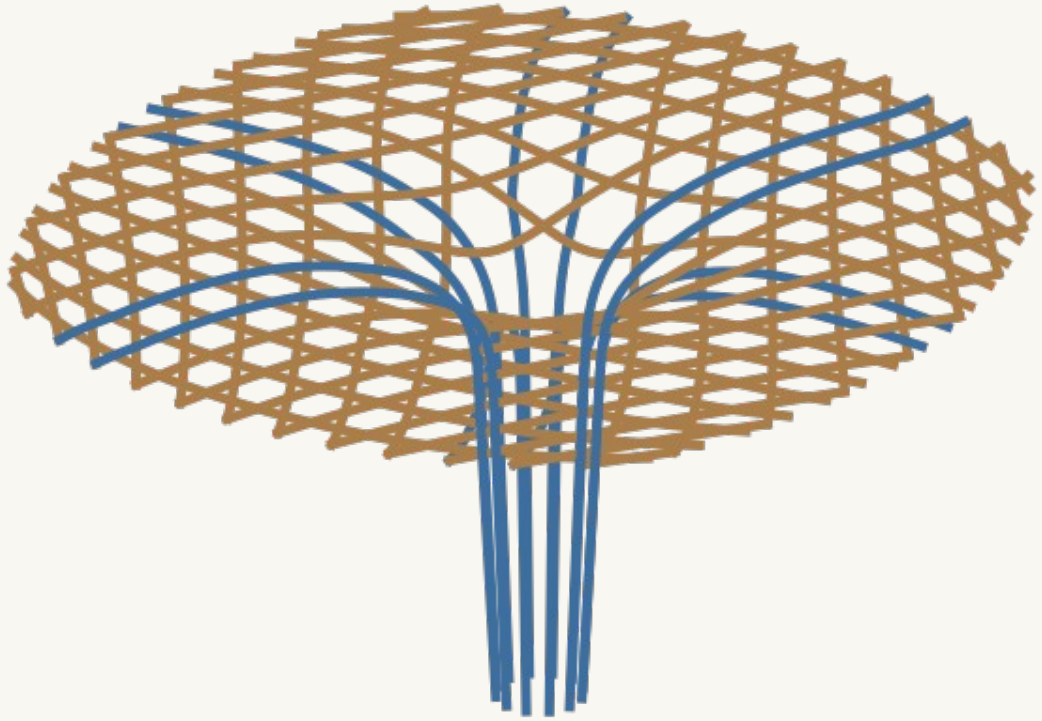
02 / 03: QUALITATIVE MECHANISMS

Shade % does not equal cooling °C.

Shade: project P2/P5. Greening mechanisms: [1,2]. Arrows are conceptual; no temperature or airflow field is simulated.

Separate the support role from the crown role

Classify members by whether their centre lines reach the ground.



ZONE	PROPOSED SECTION	LENGTH
12 steel supports	Q355B tube Ø80 × 4 mm	69.38 m
47 bamboo crown members	Engineered bamboo solid Ø80 mm	219.28 m
TOTAL	59 members	288.66 m

Steel supports address compression and wind bending. The bamboo crown is governed by stiffness, treatment and replaceability.

Project take-off and proposed member schedule (P1, P3). Sections and material mass remain preliminary specifications.

Five criteria guide the material selection

Durability · cost · thermal conductivity · carbon · environmental adaptability.

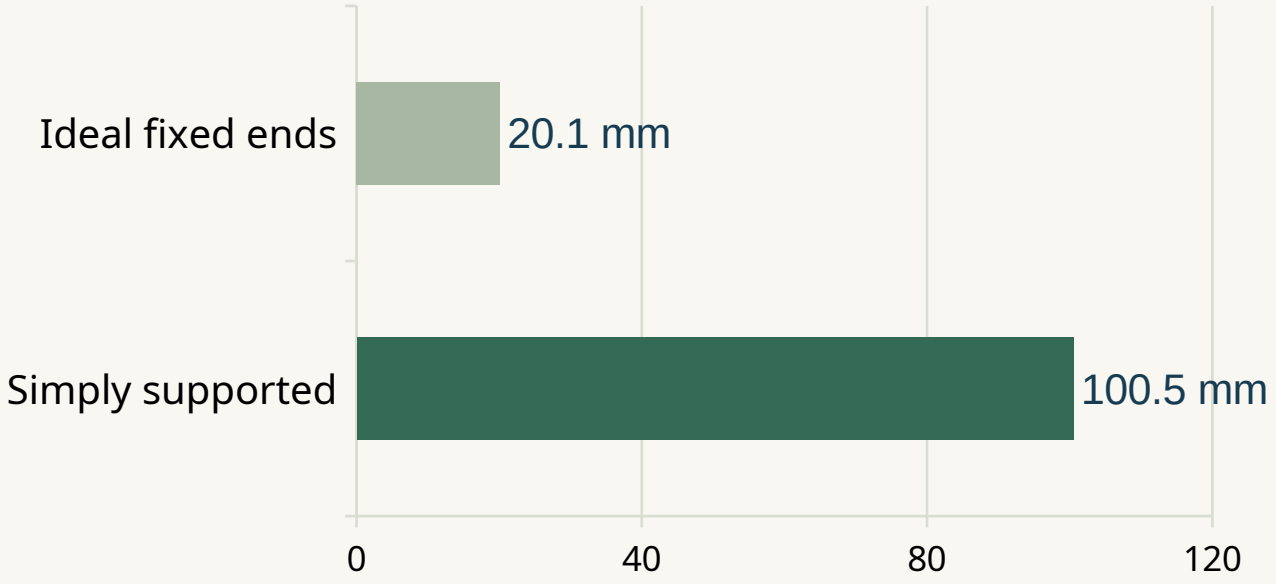
MATERIAL / SECTION	BENDING* kN·m	λ* W/(m·K)	CARBON* $kgCO_2e/kg$	MATERIAL COST* CNY / 288.66 m
Q355B tube Ø80×4	6.14	50	2.15	18,394
Q355B tube Ø80×3	4.78	50	2.15	13,983
Engineered bamboo Ø80	1.01	0.22	0.40	10,158
Treated timber Ø80	0.85	0.14	1.10	6,530
Raw bamboo Ø80	0.40	0.18	0.25	4,330

DURABILITY	ENVIRONMENTAL ADAPTABILITY
Protect steel coatings; treat bamboo; detail inspection and replacement.	Check moisture, UV, wind exposure, local supply and plant suitability.

Source: material-comparison workbook (P3). *Preliminary inputs; product, processing and factor boundaries need verification.

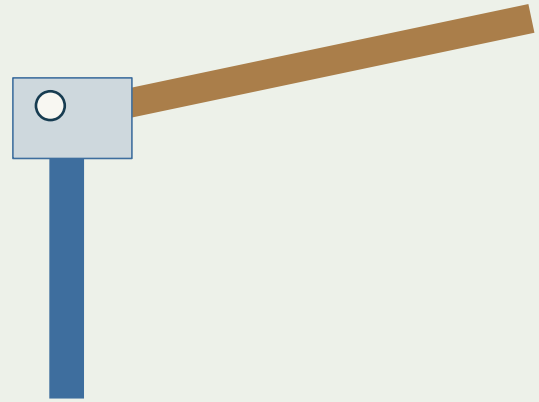
The crown is stiffness-sensitive: joints matter

Load assumption → boundary condition → deflection → required connection behaviour.



6.65 m SPAN · ADOPTED L/200 = 33.25 mm

PROPOSED SUPPORT-HEAD NODE



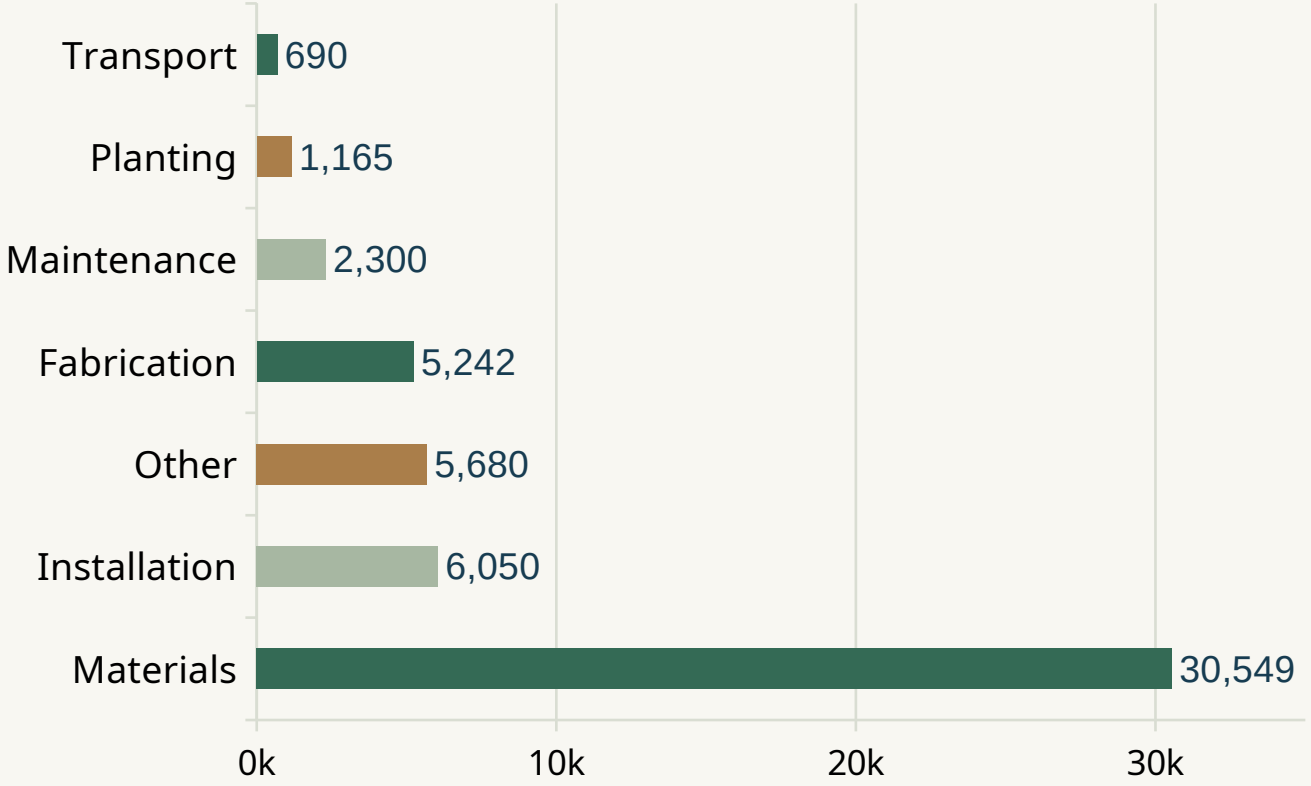
Node plate + sleeve + structural adhesive
with high-strength bolts

Concept only; rotational restraint must be verified.

Simplified beam calculation and proposed node method (P3). Wet foliage 18 kg/m² and bamboo properties are assumed inputs.

The unit budget includes fabrication, installation and care

Preliminary total: CNY 51,676 per unit—not simply the price of the members.



MATERIALS = 59.1% OF TOTAL

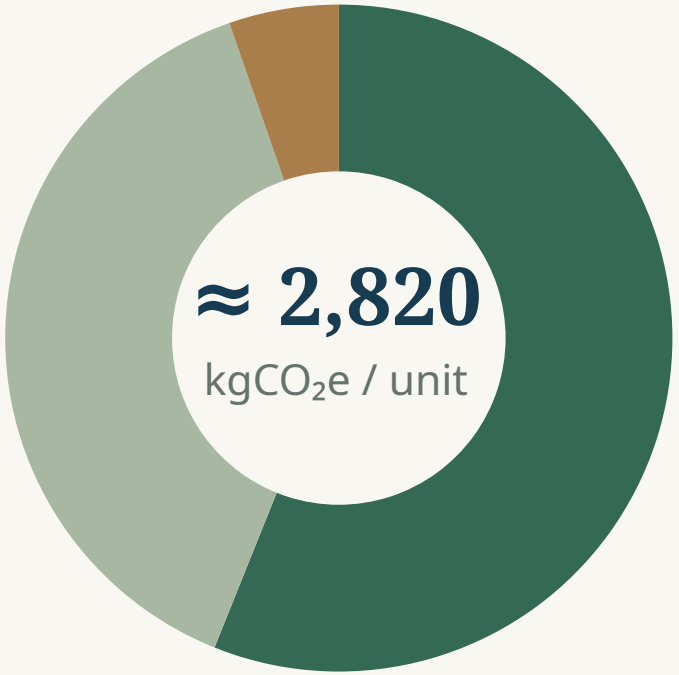
KEY COMPONENT	CNY
Bamboo crown	7,716
Steel supports	4,421
12 rigid-head nodes	12,000
Counterweight	3,402

These component rows are part of the full total. Nodes are a larger allowance than the primary steel members.

Source: Cost-Analysis.xlsx (P3). All rates remain reference estimates rather than confirmed quotations.

Account for the members, base and maintenance needs

Current carbon estimate: approximately 2,820 $kgCO_2e$ per unit.



CONTRIBUTOR	$kgCO_2e$
Primary members	1,581
Counterweight	1,089
Other components	150

Counterweight ≈ 39% of the current estimate

STEEL	BAMBOO	CONNECTIONS
Protect coating	Treat and inspect	Check fixings and grout

Workbook estimate (P3). Factor boundaries and node inclusion require checking; no biogenic-carbon credit is taken.

Keep the planting mix conditional on the site

Source proposal: 2 Boston ivy + 1 Rangoon creeper per unit; trumpet creeper is an alternative.

PRIMARY • 2 PLANTS

Boston ivy

Parthenocissus tricuspidata

Deciduous: summer leaf cover, winter leaf loss. Adhesive tendrils; train and check grip on the support.

COMPANION • 1 PLANT

Rangoon creeper

Combretum indicum

Warm-climate climber. Check winter survival before adoption; provide guides and routine tying.

ALTERNATIVE

Trumpet creeper

*Campsis grandiflora**

Vigorous woody climber; check roots, pruning demand and attachment. Confirm nursery species.

ROOTS + CARE

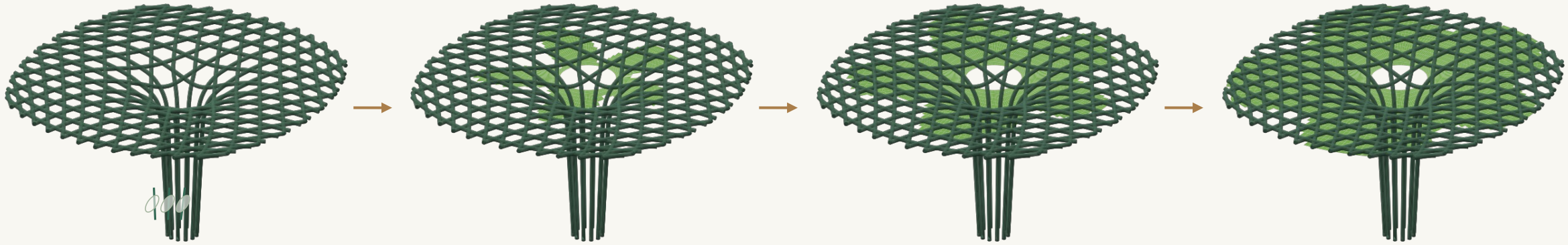
Source allowance: 1.2 m^3 substrate, drip irrigation, training and pruning.

*“Trumpet creeper” can refer to different Campsis species/hybrids. Nursery identification remains provisional.

Project planting screen P3. Botanical traits need site/nursery confirmation. No measured annual growth rate is adopted.

From young climbers to a leafy crown

Concept sequence: plant → climb → spread → maintain. Cover states are illustrative, not dated growth records.



01 PLANT & TRAIN

Young plants; guides + water

02 REACH THE CROWN

Train along the supports

03 SPREAD LATERALLY

Branch across crown members

04 MAINTAIN SHADE

Prune, irrigate and inspect

ILLUSTRATIVE ROUTE TIME	**Assume 1 / 2 / 3 m per season → about 6 / 3 / 2 growing seasons**

Route ≈6.9 m; starting shoot 1.5 m assumed; ceil[(6.9 − 1.5) / rate]. This is NOT time to full cover.

P5 spatial-cover models reused as concept images. Time calculation P6: assumed rates only; crown coverage and opening are separate checks.

Project shade onto one fixed reference zone

30°N · assumed X=east/Y=north · true solar time 09–15 · no surrounding obstructions.

01	SUN DIRECTION
02	MODEL + LEAF ENVELOPE
03	GROUND PROJECTION
04	TARGET INTERSECTION

IDEAL GEOMETRIC OVERLAP

Continuous-leaf shadow ∩ target ÷ target area

EQUIVALENT DIRECT-BEAM BLOCKAGE

$$A_{eq} = A_{bare} + (1 - \tau)(A_{leaf} - A_{overlap})$$

Divide Aeq by target area; τ = 0.15.

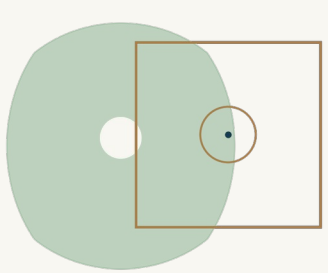
TARGET: 5 × 5 m MINUS r = 0.75 m → 23.23 m²

Not net usable floor area.

Solar geometry: NOAA [3]. Project calculations and scenario assumptions: P2, P5. Foliage/transmittance are not measured.

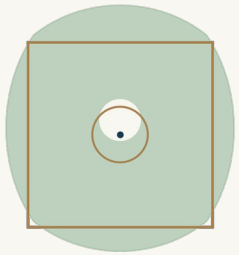
The shadow position and solar season both matter

Ideal continuous-leaf overlap: a high noon value does not mean continuous occupied-hour shade.



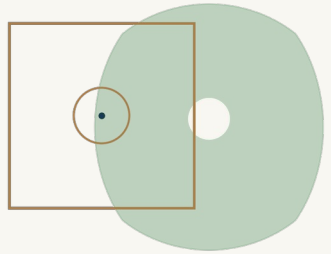
09:00

47.2%



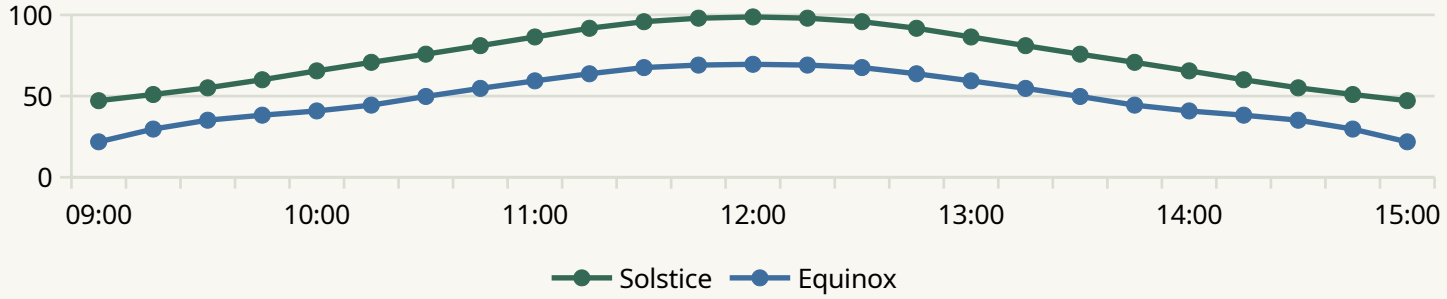
12:00

98.7%



15:00

47.2%



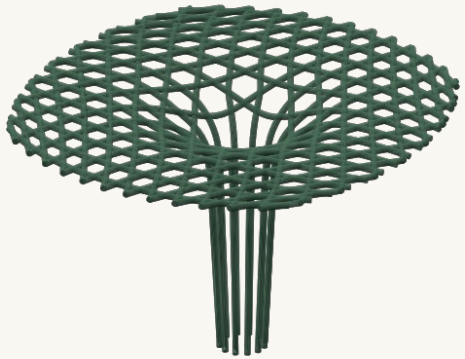
09–15 TIME MEAN

Solstice 75.3%
Equinox 49.8%

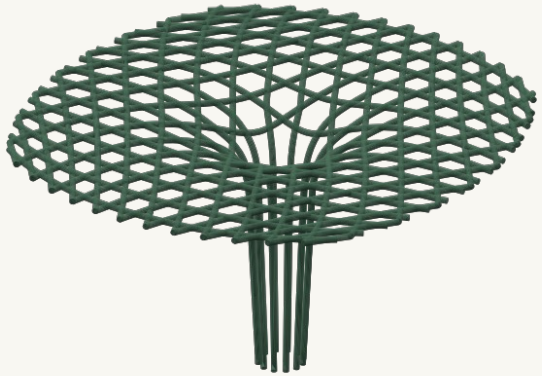
Project projection data (P2, P5). Solar-season comparison, not a calibrated seasonal plant-growth model.

Seven models test three geometry controls

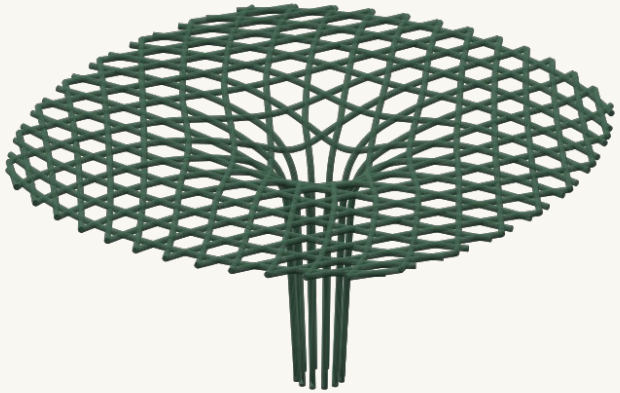
One variable at a time; same target, 50% plan cover, solstice solar hours and τ = 0.15.



SMALLER CROWN



BASELINE



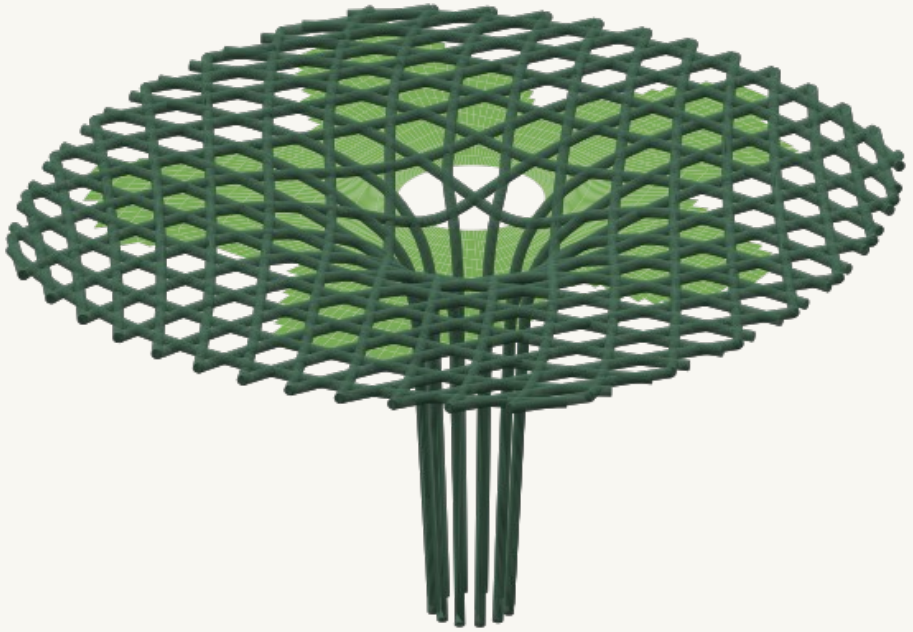
LARGER CROWN

CONTROL	TESTED SETTINGS	MEAN BLOCKAGE: LOW / BASE / HIGH
Upper expansion	0.92 / 1.08 / 1.24	52.3% / 60.2% / 66.1%
Upper-height control	−0.4 / 0 / +0.4 m	62.6% / 60.2% / 57.8%
Corner factor	0.55 / 0.85 / 1.00	61.8% / 60.2% / 59.1%

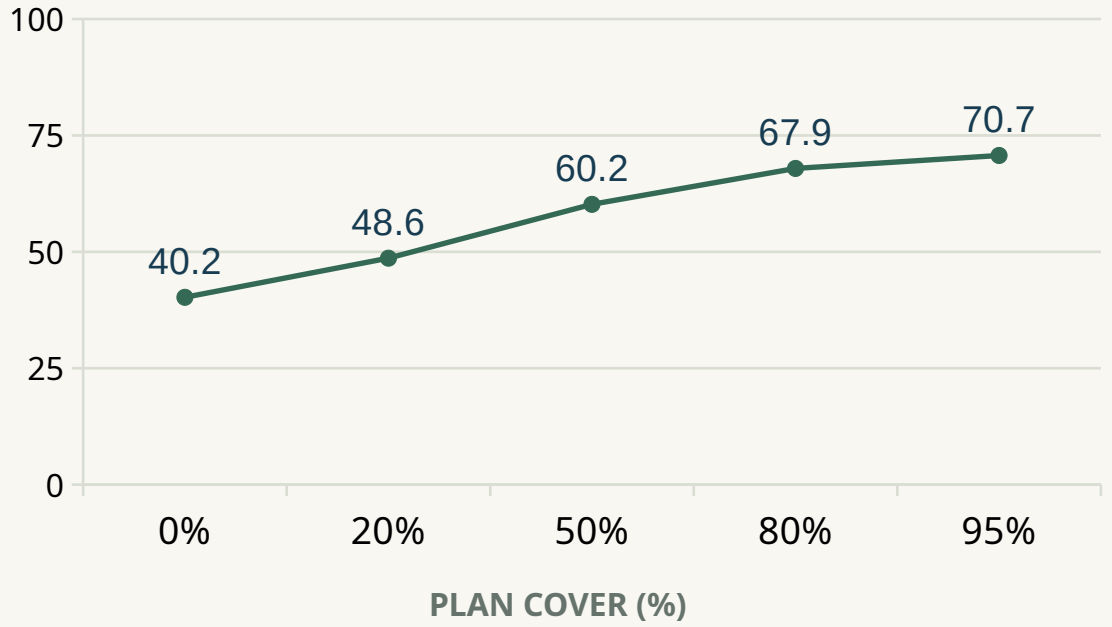
Project parameter study (P5). Controls are not uniform scaling; larger cover percentage does not imply equal plant/material input.

Coverage amount alone does not determine useful shade

Fixed final structure; five nested leaf-cluster scenarios, not observed growing months.



STATIC VIEW: 50% PLAN COVER • ANIMATION IN PPT

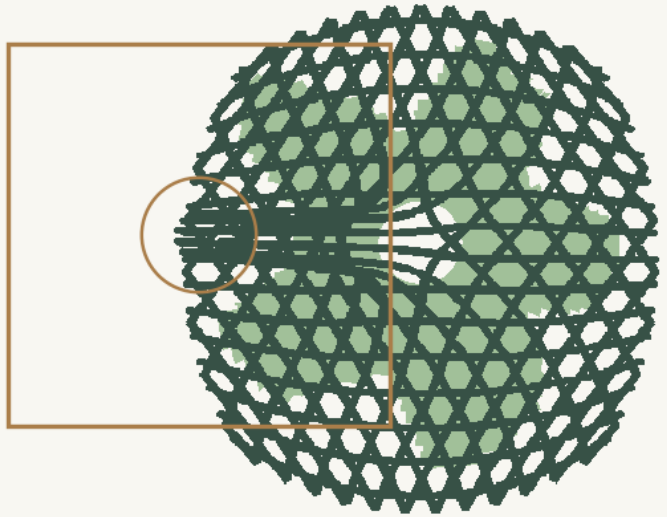


Same 50% cover, different layouts at noon: 77.5% / 71.4% / 62.6% blockage.

Project spatial-cover calculation (P5). Frame/leaf overlap deducted, τ = 0.15; equivalent direct-beam blockage, not cooling.

Put occupied-hour shade into the layout decision

A rest node must coordinate seating, through movement, roots and a substantial base.



STATIC VIEW: 15:00 • FIXED 50% PLAN COVER

01 SEATING LOCATION

Use the occupied-hour footprint, not just the crown outline or shadow centroid.

02 SPACE FOR ACCESS & ROOTS

Allow for circulation, counterweight, root zone, irrigation and maintenance.

03 VALIDATE THE OUTCOME

Confirm site and planting; compare radiation, temperature, humidity and wind.

Conceptual application P4; fixed-50%-cover projection P5. No real-site compliance or measured comfort result is claimed; method lead [4].

Coordinate prefabricated modules and site interfaces

Numbered components help assembly; long members, heavy base and grouted joints constrain it.

MODULE	WHAT IS PREPARED
Crown	47 formed/treatment-ready bamboo members
Supports	12 proposed galvanised steel members
Connections	Head nodes, sleeves, clamps and base flange
Foundation	Precast counterweight with anchorage
Planting	2 + 1 plants, substrate and drip provision

01 **Prefab, treat & number**

02 **Lift base and set out**

03 **Support, align & assemble**

04 **Grout, cure & recheck**

05 **Plant, train & restore**

14 person-days estimated ≠ 14 calendar days. The ≈9.07 t base controls lifting; grouted nodes limit lossless reversal.

Project member/assembly and cost schedule (P3, P4). Proposed detail and labour quantities, not a tested construction programme.

Connect geometry, construction and useful shade

Three design workstreams now have traceable outputs.

GEOMETRY

Rhino/GH model and parameters

Final form, drawings and material take-off

MATERIALS & BUILD-UP

Screening, sections, budget and modules

Proposed interfaces and implementation constraints

PERFORMANCE

Time-dependent shade and sensitivity

Geometry, cover and spatial-distribution effects

7 PARAMETER MODELS

5 COVER STATES

1,750 TIME-STEP RESULTS

Next: confirm products and joints, calibrate plant establishment, and test the real site.

Synthesis of project records P1–P5. Product, joint, site and growth-duration evidence are the next verification priorities.

References and project evidence

Method sources and project inputs are separate; drawings, brief and schedules accompany the deck.

[1] OUTDOOR SHADE & COMFORT

Watanabe et al. (2014). Evaluation of outdoor thermal comfort in sunlight, building shade, and pergola shade during summer in a humid subtropical region. Building and Environment, 82, 556–565.

[2] GREENING MECHANISMS

Hoelscher et al. (2016). Quantifying cooling effects of facade greening: Shading, transpiration and insulation. Energy and Buildings, 114, 283–290.

[3] SOLAR GEOMETRY

NOAA Global Monitoring Division. General Solar Position Calculations.
gml.noaa.gov/grad/solcalc/solareqns.PDF

[4] COMFORT METHOD

UTCI Calculator / COST Action 730.
www.utci.org/utci_calc.php
Method lead only; no project UTCI result is reported.

PROJECT RECORDS

P1 Model & take-off • P2 Ideal shade analysis • P3 Cost-Analysis.xlsx
P4 Proposed design & build-up • P5 Parameter and spatial-cover simulation

P6 Plant establishment concept and assumed-rate route sensitivity; not a measured growth forecast.

Literature entries were supplied with project records; original-source checks remain pending. Unverified factors are not presented as certified standards.